AMUDIESHWAR A G

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LinkedIn | GitHub | Portfolio

Professional Summary

Pre-final-year B.Tech Artificial Intelligence and Data Science student with hands-on experience in AI/ML, Generative AI, RAG pipelines and full-stack application development. Interested in building practical technology solutions using machine learning and modern software technologies.

Education

Rajalakshmi Institute of Technology, Chennai — Expected 2028
B.Tech – Artificial Intelligence and Data Science — CGPA: 8.41/10
Higher Secondary Education: **Montfort Matric Hr. Sec. School, Tindivanam** — 92%

Technical Skills

Programming: Python, Java, C, SQL
AI/ML: Machine Learning, Deep Learning, XGBoost
Generative AI: RAG
Data Analysis: Pandas, NumPy, Matplotlib
Application Development: React.js, Flutter, TypeScript, FastAPI, Tailwind CSS
Databases: PostgreSQL, Supabase, MySQL, Firebase
Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Projects

DeviceGuardian AI — *FastAPI, Python, XGBoost, SHAP, Flutter, Kotlin*

- Built a cross-platform predictive maintenance system to monitor laptop and smartphone telemetry and identify potential hardware degradation.
- Implemented XGBoost predictions and SHAP-based explainability using synthetic telemetry generated through a NASA-informed methodology.

LexGuard AI — *FastAPI, React, TypeScript, Gemini API, FAISS*

- Developed a legal document analysis platform to identify contract liabilities, indemnities and termination risks.
- Built an asynchronous RAG pipeline using PyMuPDF, FAISS and Google Gemini to analyze clauses and generate due diligence PDF reports.

ExpenseIQ — *Flutter, Dart, Supabase, PostgreSQL*

- Developed a cross-platform expense tracker with cloud synchronization, budget monitoring, overspend alerts and interactive analytics.
- Implemented CSV exports, browser-compatible downloads and Supabase PostgreSQL integration with Row Level Security.

HireConnect — *React, TypeScript, Firebase, Tailwind CSS, PWA*

- Developed a bilingual Tamil-English recruitment platform connecting degree holders and vocational workers with employers across Tamil Nadu.
- Implemented qualification-based scoring, direct-call hiring and Service Worker caching for improved accessibility in low-bandwidth environments.

AI Stroke Detection — *PyTorch, EfficientNet-B0, Grad-CAM, FastAPI, React*

- Developed a deep learning application to classify head CT scans into ischemic stroke, hemorrhagic stroke and normal categories.
- Integrated EfficientNet-B0 and Grad-CAM with FastAPI to display model-generated visual heatmaps in a React interface.

Internship Experience

Data Science Intern — 01 - 31 July 2025
CodSoft – Virtual Internship

- Completed a four-week internship; independently developed Titanic Survival Prediction and Movie Rating Prediction projects.

Emerging Technologies (AI & Cloud) Intern — 14 July 2025 – 6 August 2025
Edunet Foundation, AICTE and IBM SkillsBuild

- Completed practical training in AI and Cloud technologies; explored IBM watsonx.ai and Granite and developed a Generative AI Recipe Preparation Agent.

Achievements & Leadership

- **Robochipx Hackathon – First Place, AI Track:** Led Team System Hackers to first place among 100 competing AI teams in a 24-hour hackathon; won INR 10,000.
- **Smart India Hackathon 2026 – Internal Selection:** Led Team System Hackers through RIT's institutional selection for problem statement SIH26145 (Blockchain and Cybersecurity).
- **MakeGPT Summit 2025 – IIT Madras:** Represented RIT's AI and Data Science delegation at IIT Madras Research Park.

Certifications

- Retrieval Augmented Generation with LangChain – IBM SkillsBuild
- Journey to Cloud: Envisioning Your Solution – IBM SkillsBuild
- Getting Started with Artificial Intelligence – IBM SkillsBuild